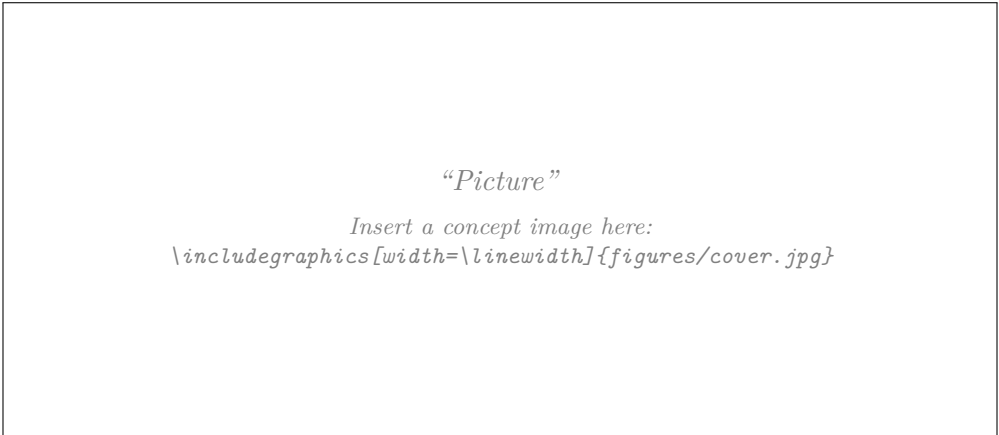


Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City



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Comments	[TODO: optional remarks for the instructors]

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1 Introduction

1.1 Background and relevance

Impervious urban surfaces absorb most of the incoming short-wave radiation and store it, sealing removes the evaporative cooling that vegetation would provide, and the narrow geometry of a street canyon keeps the emitted long-wave radiation from escaping to the sky (Oke 1982). Zurich measures the result as an air-temperature difference of 1 K to 2 K by day and of up to 7 K at night against the surrounding area, with sun-exposed sealed surfaces above 60 °C in summer (Stadt Zürich 2021). The stored energy is released into the street canyon, and that is where the load is felt: a surface at 60 °C radiates at eye level onto anyone walking, waiting or sitting in the street, while the same person indoors is shielded from it and can ventilate at night. Public outdoor space is the part of the city that cannot be conditioned, and it is where the city's health concern applies — cerebrovascular, cardiovascular and respiratory illness during heat periods, with older people and small children most at risk (Stadt Zürich 2021).

Air conditioning is no answer to this: it rejects the heat into the street as waste heat and raises electricity demand at the peak hour, which is why the city advises against it (Stadt Zürich 2020a, pp. 132–133). What Zurich pursues instead — the *Fachplanung Hitzeminderung* with its thematic maps and implementation agenda (Stadt Zürich 2020a; Stadt Zürich 2020b; Stadt Zürich 2020c) — acts on the outdoor space at pedestrian level. That space is bounded by pavement and by the façades of a building stock that already stands, so those surfaces are the intervention point a retrofit can reach: the target is the street in front of the façade, not the room behind it. Section 2.1 sets out what the city specifies for them today, and with it the benchmark this project has to beat.

1.2 Aim of the report and intended contribution

This report documents a biomimicry design process for reducing heat load on the envelope of existing urban buildings, and with it on the street space those envelopes enclose. It follows the Biomimicry Design Spiral (Baumeister et al. 2014) through scoping, biological discovery and abstraction, engineering concept development, validation and evaluation, in the challenge-to-biology direction described by Helms et al. (2009). The intended contribution is a technically specified, quantitatively checked envelope concept, together with a transparent account of which biological mechanisms were transferred, which were rejected, and on what evidence.

1.3 Structure of the report

This report is organised as the reasoning chain it documents, and it is written so that the chain can be followed backwards as well as forwards: challenge → function → biology → abstraction → concept → engineering check → evaluation → iteration.

Section 2 cuts the given challenge down to something answerable. It fixes a reference city and a reference site, states the design question and the four candidates rejected to reach it, draws the system boundary, lists the functions as solution-neutral verbs and turns them into measurable criteria. Section 3 carries out the biological search those functions define, abstracts the mechanisms found into transferable principles, develops the concepts, and tests them. Section 4 assesses the result against Life's Principles rather than listing them, and Section 5 asks what the biomimicry route contributed that a conventional one would not have, and where the evidence runs out.

Because the method is judged as much as the result, **every substantive decision in this report is stated together with the alternative it displaced and the reason.** Those decisions are argued where they arise, and collected in one place in Appendix B, which can be read as a map of how the project reached its current shape.

2 Challenge & Scoping

2.1 Problem definition and design question

Given challenge (Challenge 1). *Cities are facing increasing heat stress due to climate change. Develop bioinspired strategies to reduce urban heat through building materials, façade design, or urban planning.*

The challenge as given spans three scales — material, building component and urban planning. Working on all three would make the project larger rather than more focused, which the systems view is explicitly meant to prevent. The design question is the instrument that makes this cut.

2.1.1 Candidate questions considered

Four candidates were formulated in the team and assessed against the criterion that a design question must be neither too broad nor too narrow: open enough to admit different solutions, but specific enough to say what is to be reduced.

Table 1: Design questions considered, and why none was adopted unchanged.

#	Candidate	Assessment
1	How do we prevent heat build-up in urban areas?	<i>Pre-commits to prevention</i> , which excludes the removal half of the problem — the half the incumbent measure leaves unsolved. Also unanchored in place and time.
2	How could we use the energy or heat build-up positively while lowering temperature?	<i>Unanchored</i> . No location, and “positively” does not say for what.
3	How might we get the heat that Zurich’s existing buildings store during the day back out of the city before the next morning, without using energy?	Well formed and highly testable , but it fixes the mechanism (night removal) and the constraint (no energy) before the scoping has done its work. Its self-regulating intent is carried forward as a design decision.
4	How can we prevent heat waves causing heat islands?	<i>Factually wrong</i> . Heat waves do not cause heat islands. The urban heat island is a permanent consequence of urban surface properties and geometry and exists on ordinary summer days; a heat wave makes it dangerous, it does not create it (Oke 1982).

2.1.2 Selected design question

Design question (solution-neutral):

How might we reduce the urban heat island effect in cities like Zurich?

The question is stated at the level of the *effect*, not of the intervention. It names no solution — no coating, no greening, no geometry, no water — and, unlike candidate 1, it does not pre-commit to *preventing* heat build-up, which would exclude the removal half of the problem. That exclusion would matter: the city’s own modelling puts the effect of a bright façade at -1.0 K by day but only -0.1 K at night (Stadt Zürich 2020a, Tab. 11, 128), against a heat island that reaches 7 K after dark (Stadt Zürich 2021). The night is the larger half of the problem and the half the incumbent measure does not address.

What the question deliberately does not do is narrow the project; that is the job of the rest of this section, and each step is argued rather than assumed. Section 2.2 fixes the reference city, the reference site and the place the effect must appear, which makes the **façade** the system of interest; Section 2.3 fixes what the design may touch; Section 2.4 states what it must do; and the criteria in Section 2.6 fix what “reduce” has to mean in numbers.

One property is carried forward from candidate 3 and is not visible in the wording of the question, so it is stated here explicitly: the response must be **self-regulating**, its cooling power rising with the heat load instead of being fixed. This is a decision of the team rather than a demand of the question, and it is the decision that separates the project from the incumbent measure. A bright surface has a fixed optical response: it reflects the same fraction of radiation at 20°C as at 40°C , and it does nothing at all once the sun has set. A self-regulating surface behaves like a biological one — a leaf’s response is driven by the same solar energy that heats the leaf, so the cooling effort scales with the load, without a controller, a sensor or a pump. It becomes function F7 and is measured by the first performance criterion.

That decision also resolves the team’s initial difficulty of having two systems of interest (“reduce heat” and “prevent heat”). These are not two systems. They are two functions of one system: prevention acts during the day window, removal during the night window, and the self-regulating link connects them.

2.2 Context and constraints

Reference city: Zurich. All context values, benchmarks and engineering checks in this report refer to the City of Zurich. Fixing one reference city is what makes the constraints and the criteria checkable instead of generic. Zurich was chosen for four reasons: it has an adopted municipal heat-mitigation strategy, the *Fachplanung Hitzeminderung* (Stadt Zürich 2020a), with an implementation agenda (Stadt Zürich 2020c), a guidance leaflet for building owners (Stadt Zürich 2021) and three thematic maps (Stadt Zürich 2020b), so the project has a real client, a documented baseline and published target values instead of invented ones; it publishes an open climate analysis and open geodata (Stadt Zürich 2026), so the reference site can be selected from measured heat-load data rather than intuition; MeteoSwiss operates long-term stations in and around the city (MeteoSwiss n.d.), which supplies the boundary conditions (G , T_a , wind speed, humidity) for the energy balance in Section 3.2; and its building stock and morphology are representative of the Swiss Mittelland, so results transfer to comparable Swiss cities.

Reference site and users. We adopt the city’s own model area *Modellierungsgebiet 03 — Geschlossene Randbebauung* (closed perimeter block) as the reference site (Stadt Zürich 2020a, pp. 148–151). It consists of two outward-closed blocks of five storeys plus a roof structure enclosing private courtyards, in a dense district close to the centre; the street space is sealed from façade to façade and only isolated trees provide shade. The city’s own assessment is that the reduced air exchange and the lack of cold-air supply make this structural type overheat

comparatively quickly. Choosing this site has a methodical advantage: the Fachplanung already publishes modelled baseline and climate-optimised results for exactly this area, so our concept can be compared against a documented reference rather than against our own assumptions. **[TODO: Record address, façade orientations, roof area and year of construction of the specific building you pick inside this area.]**

The users are the residents of the upper floors, who carry the strongest heat load and usually have no mechanical cooling; the pedestrians in the street canyon, who receive radiation from the façade at eye level; the building owner, who decides on and pays for the retrofit; and the trades who install and later maintain the system.

The design case. The design case is not a single hot afternoon but a **multi-day heat wave**: five consecutive hot days without rain. Within each day, three windows matter and they follow one another without a gap, because the surface never stops exchanging heat — the *load* window around solar noon, when the energy that has to be dealt with all night enters; the *evening* from 18:00, when people are outdoors and west façades reach their highest incidence angle; and the *night* to 06:00, when the store must be discharged before the next day begins. A concept that works on day one and is saturated by day three has not solved the problem. **[TODO: Confirm the duration and frequency of such spells for Zurich from MeteoSwiss records, and state the station and reference period.]**

Constraints. Table 2 separates hard constraints (a violation disqualifies a concept) from soft ones (they weight the evaluation in Section 3.3). The constraints on installation and maintenance come from the team’s observation that the people who mount and service the system are part of the context; they are recorded here rather than as subsystems, because the system of interest contains components, not people.

2.3 System view and project boundary

A system view is a deliberate change of altitude. Before deciding what to build, we ask what the object sits inside, what sits inside it, and what else solves a comparable problem next to it. The purpose is not tidiness: the level *above* shows which effects we can influence and which we merely inherit, the level *below* shows where a design decision actually lands, and the systems *beside* show which resources and constraints already exist. The boundary drawn afterwards then does the opposite job — it keeps the project finishable by naming what we will not touch. Table 3 lists the levels and Figure 1 draws them with the boundary in place.

Read downwards, the table says the following. The **super-system** is the city: its microclimate, the geometry of the street canyon and the radiation balance between the two are what produce the heat island, and they are the conditions our design meets rather than quantities it can set. The **system of interest** is deliberately one layer and not a building — the outer skin is the surface where the city’s radiation actually arrives and where the building’s stored heat is actually released, so it is the thinnest object that still contains the whole problem. The **parallel systems** are the three that already act on the same street: the pavement, treated by the same client with the same high-albedo logic, which sets a benchmark our envelope has to beat; the power grid and building energy system, which is the route a conventional answer would take and which the energy constraint closes to us; and the residents, who fix *where* the effect has to appear — at head height in the canyon, not on the roof membrane. The **subsystems** are what a concept may actually vary, and they are components rather than people: the optical outer surface, the layer behind it that stores and insulates, the materials both are made of, and the fixing to the wall.

Table 2: Context conditions and constraints.

Type	Class	Constraint
Energy	hard	Limit auxiliary electrical energy for the cooling function. The heat load, not a supply, must drive the response.
Water	hard	No water waste.
Existing building	soft	Retrofit onto an existing envelope is preferable, but not necessary.
Townscape	hard	The result must be acceptable in the Zurich townscape. Buildings may be listed; heritage protection, building law and neighbour rights can restrict visible changes (Stadt Zürich 2020a, p. 128).
Glare	hard	No specular glare towards streets, neighbouring flats or traffic. The city names glare as the main objection to bright envelopes (Stadt Zürich 2020a, p. 128), which favours diffuse over mirror-like surfaces.
Climate	hard	Must survive the Swiss annual cycle: frost to about -10°C , freeze–thaw cycles, driving rain, UV, and periods of several dry weeks in summer.
Winter	hard	Must not appreciably increase heating demand. A high solar reflectance also reduces useful winter solar gains; this trade-off must be quantified, not ignored.
Installation	soft	Must be installable by ordinary trades on an occupied building, ideally without full scaffolding.
Maintenance	soft	Must remain functional with little maintenance. Soiling and the resulting cleaning cost are named by the city as a specific weakness of high-albedo envelopes (Stadt Zürich 2020a, p. 128). Service life target ≥ 20 years.
Solar energy	soft	Should not unnecessarily compromise roof- or façade-integrated photo-voltaics; the city notes that bright envelopes reduce PV yield (Stadt Zürich 2020a, p. 128).
Structural	soft	Limited additional dead load and build-up thickness on an existing envelope.
Embodied emissions	soft	The greenhouse gas embodied in the added build-up must be repaid by the avoided cooling demand within the service life, measured in $\text{kg CO}_2\text{-eq l/m}^2$. [TODO: source a benchmark]
Economic	soft	The city expects higher cost for high-albedo envelopes (Stadt Zürich 2020a, p. 128). Target: cost per square metre in the range of a normal refurbishment.

Table 3: System hierarchy.

Level	In this project
Super-system	The city and its existing infrastructure
System of interest	The heat-regulating outer layer of a building
Parallel systems	Stormwater and sewer network · street and pavement network · power grid and building energy system · residents, transport and public transit
Subsystems	Outer surface · storage and insulation layer · materials · attachment to the substrate

Table 4: Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Prevent heat build-up on, and remove heat from, the building façade	Redesign of the building itself. Reuse of the stored energy. Internal cooling.

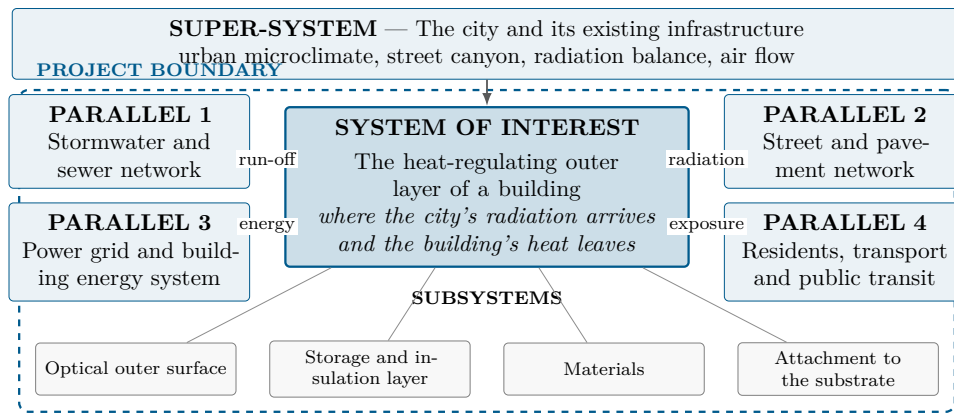


Figure 1: System view. The dashed line is the project boundary: everything inside it is designed and analysed by the team, and the parallel systems sit deliberately outside it. The subsystems are components and functions, not people — the trades who install and maintain the system appear as constraints in Table 2.

2.4 Functions and functional diagram

Functions are what the design must *do*, written as verbs and kept free of any solution. They are stated at the scale of the designed object, not at the scale of the city: the design regulates heat at the surface it occupies, and the city-scale effect is the *goal*, not the function — in the same way that the function of polar bear fur is to insulate, not to make the Arctic habitable.

Table 5: Function list. Main function, sub-functions and supporting function. Every entry is used later: by a performance criterion in Section 2.6, by a row of the morphological matrix in Section 3.2, or by both.

ID	Function (verb)	Explanation / taxonomy link
F1	Regulate heat gain and release at an urban surface	Main function. Modify / regulate physical state: manage temperature
F2	Reflect short-wave radiation	Prevention — avoid absorbing solar energy in the first place
F3	Shade a surface	Prevention — reduce the irradiated area, e.g. through surface geometry
F4	Emit long-wave radiation	Removal — release heat to the sky, in particular in the atmospheric window
F5	Transfer heat to ambient air	Removal — convective dissipation at the surface
F6	Store thermal energy temporarily	Removal — buffer the daily peak and shift it into the night window
F7	Couple response strength to load	The core of the project. Increase removal (F4, F5) as surface temperature rises, without external control
F8	Resist degradation	Supporting — withstand UV, frost, driving rain and soiling over the service life

What the functional diagram shows. A functional diagram is the function list drawn as a flow picture. It contains only *functions* and the *flows* between them — energy, material and information — and deliberately no components and no materials, so that it stays solution-neutral: the same diagram must remain true whichever concept is later chosen. Reading Figure 2 left to right, solar radiation arrives and is either turned away (F2, F3) or absorbed and buffered (F6); what is stored leaves again by radiation to the sky (F4) or by convection into the canyon air (F5); and whatever neither path removes is the residual flow into the building, which the

design exists to minimise. The dashed orange arrow is F7 and is the only loop in the picture: the surface temperature itself feeds back into how strongly the removal functions act.

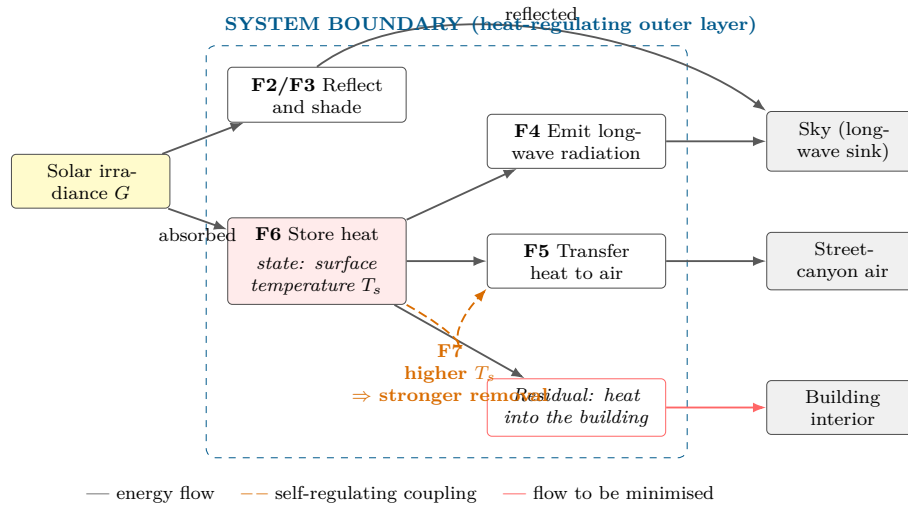


Figure 2: Solution-neutral functional diagram. Only functions and flows are shown, no components and no materials. The orange coupling is F7 and is what distinguishes this project from a fixed high-albedo surface: the state variable T_s feeds back into the strength of the removal functions. The supporting function F8 acts on everything inside the boundary and is omitted for readability.

Why F7 is physically plausible. Every removal path already depends on surface temperature, but not equally strongly. Differentiated with respect to T_s , the self-regulation gradients are, in $\text{W}/(\text{m}^2 \text{K})$, radiative $4\epsilon\sigma T_s^3 \approx 6$ and convective $h_c \approx 10$. The convective path is the steeper of the two, and it is also the one a surface can influence through geometry — which is why boundary-layer texture and self-shading relief, rather than a material switch alone, are the candidates carried into Section 3.2. **[TODO: Evaluate both derivatives with your own assumptions and state the ranking; this is the check that decides which mechanism carries F7.]**

2.5 Biologized questions

The functions are translated into questions to nature, which form the search strategy for Section 3.1. Each question names the function it searches for, so that the biology found in Section 3.1 can be traced back to a requirement rather than collected for its own interest. The challenge brief also asks directly what can be learned from desert plants and animals about passive cooling, and the answer shapes where we looked: the organisms most challenged by our problem, and unfazed by it, live where solar load is extreme and water is scarce.

- **F1** — How does nature use the heat it receives to get rid of that same heat?
- **F2** — How does nature reflect intense solar radiation without pigments that bleach, and without a mirror finish?
- **F3** — How does nature shade a surface, even in the desert?
- **F4** — How does nature release excess heat to the night sky?
- **F5** — How does nature move heat into the air around it?
- **F6** — How does nature buffer a daily temperature extreme instead of resisting it?
- **F7** — How does nature make a response stronger when the load is stronger, without a controller?
- **F8** — How does nature keep a functional surface clean without cleaning it?

2.6 Performance criteria and KPIs

The criteria are derived from the functions, interfaces and constraints identified above, not invented independently of them. They are used twice: to guide the engineering checks in Section 3.2 and to compare the concepts in Section 3.3. The reference case throughout is an existing Zurich façade with a solar reflectance of about 0.3, the value the city uses for un-greened façades (Stadt Zürich 2020a, p. 114). Each criterion also names the Life’s Principle it carries, so that the principles are scored with the numbers rather than listed beside them; Section 4 then assesses how well the concept achieves them and where it falls short.

1 — Self-regulation gradient (F7). The rise in cooling power per kelvin of surface temperature, $\partial P_{\text{cool}}/\partial T_s$, in $\text{W}/(\text{m}^2 \text{K})$. It must be clearly positive and larger than for a fixed high-albedo surface. This is the defining criterion of the project: it is the quantity that separates a self-regulating envelope from a bright one, and no other criterion can substitute for it. It carries *be locally attuned and responsive* — the signal, the antenna that detects it and the response must match (Baumeister et al. 2014). **[TODO: Set a numeric target once the derivative check in Section 3.2 is done.]**

2 — Solar reflectance (F2). Solar reflectance α_{sol} over 300 nm to 2500 nm, dimensionless. Target ≥ 0.80 , the threshold the city itself defines for “high albedo” (Stadt Zürich 2020a, p. 128), against a baseline of 0.3. Ultra-white laboratory paints reach about 0.98 (Li et al. 2021), and we deliberately do not aim there: the umbrella principle *create conditions conducive to life* asks for a design *optimised rather than maximised*, and past the city’s threshold further reflectance is bought with glare, lost winter gains and reduced photovoltaic yield.

3 — Peak surface temperature reduction (F1). The drop in peak surface temperature ΔT_s against the reference façade, clear sky, 14:00, in kelvin. Target $\geq 15 \text{ K}$. **[TODO: Confirm by calculation or measurement before claiming it.]**

4 — Effect on people (F1). The reduction in physiologically equivalent temperature at 2 m above ground at 14:00, in kelvin — the metric the client itself uses. The binding benchmark is the -1.0 K median that the city models for a bright *façade*; the -1.2 K it models for a bright roof is context, not our target, because the system of interest is the façade (Stadt Zürich 2020a, Tab. 11, 128). Criteria 3 and 4 must not be confused: a high-reflectance surface can be 20 K cooler than a dark one while the effect on district air temperature is smaller by roughly two orders of magnitude (Santamouris 2014; Akbari et al. 2009). Conflating them is the most common error in urban-heat-island projects, so they are reported separately throughout.

5 — Night-time removal (F4, F6). Net cooling power at 02:00 under a clear sky, in W/m^2 , together with the heat discharged across the evening and night windows, 18:00–06:00, in Wh/m^2 . Target: at least $40 \text{ W}/\text{m}^2$ instantaneous, the order of magnitude demonstrated for passive radiative cooling surfaces (Raman et al. 2014; Zhai et al. 2017), and a night-time air temperature effect better than the -0.1 K of the incumbent (Stadt Zürich 2020a, Tab. 11, 128). A second condition applies across the five-day design case rather than within a day: the surface temperature and the state of charge of the store at 06:00 must not drift upwards from one day to the next. A concept that cools well on the first day and saturates by the third fails the design case even if every single-day number is met. This criterion carries *adapt to changing conditions*, sub-principle *adapt to temporal change*.

6 — Durability of the function (F8). Retention of the cooling function after soiling and weathering, in per cent of the initial value. Target $\geq 90\%$ after three years. Soiling is the

weakness the city names for high-albedo envelopes (Stadt Zürich 2020a, p. 128), and aged cool roofs commonly lose a significant fraction of their initial reflectance (Santamouris 2014). It carries *maintain integrity through self-renewal*, and it is the criterion that rules a chemical answer to soiling out: *use life-friendly chemistry* forbids a biocide whose breakdown products leave the façade for the street below.

Note on the targets. The targets above are aspirational benchmarks set during scoping. They are hypotheses, not results. Each is checked in Section 3.2 and either confirmed, revised or explicitly declared unverified. **[TODO: Before hand-in: verify every cited benchmark value against its primary source.]**

3 Bio-Inspired Design Process

The chain this section has to make visible is challenge → function → biology → abstraction → concept → engineering check → evaluation → iteration. Section 2 delivered the first two links. This section delivers the biology and the abstraction, and hands a set of transferable principles to the concept development in Section 3.2.

3.1 Discovering & Abstracting

3.1.1 Research path

Every search started from a verb in the function list of Table 5, never from an organism. This follows the rule that biological content is organised by the function it serves, which is also the structure of the Biomimicry Taxonomy and therefore of AskNature (AskNature 2026a). Table 6 records which taxonomy branch was searched for which of our functions, and what it returned; the biologized questions themselves are in Section 2.5.

Table 6: Research path. Each of our functions was translated into a taxonomy branch before any organism was considered.

Function	Taxonomy branch searched	Models it returned
F2, F3	Modify → physical state → light/colour; Protect from non-living threats → light	Desert snail, <i>Morpho</i> , <i>Cyphochilus</i> , tree bark, cactus ribs
F4, F5	Transform/convert energy → thermal, radiant; Distribute → energy	Elephant ears, elephant hairs, jackrabbit ears, Saharan silver ant
F6	Store → energy	Sea star, desert snail
F7	Sense signals → temperature <i>paired with</i> Process signals → respond to signals	Honeybee fanning thresholds, leaf stomata, pine cone, elephant hot spots
F8	Protect from non-living threats → dirt/solids, light, ice	Lichen, lotus leaf

F7 is worth noting: it is the only one of our functions that does not sit in a single branch of the taxonomy. It is the *pairing* of “sense signals: temperature” with “process signals: respond to signals”, which is the same construction the Life’s Principles checklist uses when it asks that “the signal it sends, the antennae it uses to detect the signal, and its response match” (Baumeister et al. 2014). That the central function of our design question has to be assembled from two branches is itself a finding: it is why searching by organism would not have produced it.

Sources, in the order they were used. AskNature strategies and innovations to find candidates (AskNature 2026a); the *Genius of Biome* report for the temperate broadleaf forest, the biome Zurich lies in, for models already translated to the built environment (Biomimicry 3.8 and HOK 2013); primary literature for the mechanisms; and commercial products where the transfer has already been made, treated as state of the art rather than as models. AskNature and AI-assisted search were used to *find* candidates only. Every mechanism claim is referenced — but not yet uniformly to a primary paper: ten of the twenty-one entries in Table A.2 still rest on an AskNature page alone. That is a stated shortfall rather than a claimed standard, and the open item below lists the replacements already identified.

The inverse question. The method also asks what nature would *not* do, and the answers narrowed the search as effectively as the positive ones. Nature would not use a mirror finish, would not run a compressor, would not pump drinking water onto a wall, and would not build one fixed response for both January and July. Three of those four are constraints in Table 2; the fourth is the seasonal problem this project does not yet solve.

Which lenses were applied. Of the five lenses in the method, two did most of the work. The *operating-condition lens* — intense solar radiation, surfaces above 60 °C, freeze–thaw, weeks without rain — led to the desert organisms. The *local lens* led to (Biomimicry 3.8 and HOK 2013), because Zurich sits in the temperate broadleaf forest biome and that report translates exactly this biome into building design. The *function lens* produced Table 6. The naturalist and ecological lenses contributed context but no model that survived selection.

3.1.2 Models found, and the selection decision

Twenty-one models were compared against one criterion, stated before the search began: a model must be transferable to a *passive surface* that survives twenty years of the Swiss annual cycle. Sixteen were selected and five were rejected. The full comparison — organism, function served, mechanism and the selection or de-selection decision for each — is Table A.2 in Appendix B; the families it produced are summarised here.

Table 7: What the search returned, by functional family. The full table with one row per model is in Appendix B.

Family	Functions	Models carried forward
Prevention	F2, F3	Desert snail, <i>Morpho</i> , <i>Cyphochilus</i> , Saharan silver ant, cactus ribs, tree bark, <i>Encelia</i>
Removal	F4, F5	Elephant ears, elephant skin hairs, jackrabbit ears, elephant hot spots
Storage and charging	F6	Sea star, desert snail
Self-regulation	F7	Honeybee fanning thresholds, leaf stomata, pine cone, jackrabbit, elephant hot spots
Keeping the surface working	F8	Lichen, lotus leaf

The five rejections, and why they matter. A search is only legible if what it discarded is on the record. The *termite mound*, the standard reference for passive cooling, was de-selected because the chimney account is contested: measurements support wind-driven transient gas exchange rather than a thermosiphon, and the building usually cited for it uses fans (Turner and Soar 2008). The *Namib desert beetle*, the most-cited water model, was de-selected because fog in

Zurich is a winter phenomenon and a heat wave brings neither rain nor fog. *Thomson's gazelle* carotid rete was de-selected as a whole-organism model — it is internal physiology — though the counter-current *arrangement* is retained as an option for the liquid loop of Section 3.2.3. *Tenrec aestivation* was de-selected because it is a behaviour, and an envelope has none. *Green façades and roofs* were placed out of scope: they are already the client's HA 09 and HA 10, so they are the baseline rather than our contribution.

Two features of the comparison are deliberate. First, the same organism appears three times — the elephant supplies ear-based transport, boundary-layer hairs and localised hot spots, and these are three different mechanisms serving three different functions, not one model counted three times. Second, the rejections above use one stated criterion rather than case-by-case taste, and it is worth stating explicitly because it explains an asymmetry in what survived: most cooling in nature is metabolic or behavioural, and an envelope has neither metabolism nor behaviour. The two models that transfer most directly — the snail shell and the cactus rib — are the two that are *structures rather than processes*.

[TODO: Before hand-in: replace the AskNature page with the primary paper wherever one exists. Ten of the twenty-one rows currently rest on an AskNature page alone, and nine of them have an obvious candidate — *each of which must be opened and checked before it is cited, because this list was assembled from search results and not from the papers*: desert snail, Schmidt-Nielsen, Taylor & Shkolnik 1971 (*J. Exp. Biol.* 55), which is also the source of the 90/95 % reflectance figures; *Morpho*, Vukusic et al. 1999 (*Proc. R. Soc. B*); cactus ribs, Lewis & Nobel 1977 (*Plant Physiol.*); elephant ears, Phillips & Heath 1992 (*Comp. Biochem. Physiol. A*); jackrabbit ears, Hill & Veghte 1976 (*Science* 194); elephant thermal windows, Weissenböck et al. 2010 (*J. Therm. Biol.*); sea star, Pincebourde, Sanford & Helmuth 2009 (*Am. Nat.* 174); honeybee thresholds, Jones et al. 2004 (*Science* 305), which also covers the patriline and 36 °C claims made in Section 3.1.3 without a source; pine cone, Dawson, Vincent & Rocca 1997 (*Nature* 390). Tree bark is the one row with no clear single paper — trace the reference on the AskNature page itself. Separately, the 23 % elephant-hair figure should move from the university news article to Myhrvold, Stone & Bou-Zeid 2012 (*PLoS ONE* 7(10)), which is where it originates.]

3.1.3 Mechanisms in depth

Two mechanisms are developed further here. They were chosen because each answers a question Section 2 left open; a third, the stomatal valve, is taken up in Section 3.2, where it becomes a vent rather than a pore.

Prevention — why reflection alone is not the strategy. The desert snail survives on sun-exposed ground at up to 50 °C without moving, without water, and without a metabolism that could help. Its shell reflects about 90 % of visible and about 95 % of near-infrared light, which is above the $\alpha_{\text{sol}} \geq 0.80$ that KPI 2 asks for and close to the ultra-white laboratory paints of Li et al. (2021) — achieved with calcium carbonate rather than a manufactured pigment. The part that matters for us is what happens next. In extreme conditions the animal seals the aperture and retracts into the second whorl, so an air space separates its tissue from the shell wall nearest the hot ground. Reflection reduces what is absorbed; the air space stops what is still absorbed from reaching what must be protected. For a retrofit this maps onto a construction that already exists: a reflective outer skin in front of a ventilated cavity. That matters for the night window, because the 7 K the city loses at night is heat released from mass that was charged during the day, and a decoupled outer skin charges far less of it.

Regulation — how a switch can behave like a controller. In a honeybee colony the temperature at which an individual worker begins to fan varies between individuals, and the variation is genetic: a colony has one queen but many fathers, so it contains patriline with different thresholds. As the nest warms, first a few bees fan, then more, then many, and the nest is held below about 36 °C with no central control at all. Each bee is a binary switch; the colony response is continuous. This resolves an objection that would otherwise disqualify every switching material considered in Section 3.2: a single threshold gives an almost infinite $\partial P/\partial T_s$ at the switching point and roughly zero either side of it, which is not the graded response KPI 1 asks for. A *distribution* of thresholds across many small units restores it. The same pattern appears in Biomimicry 3.8 and HOK (2013) as the “lots of littles” principle, arrived at independently.

[TODO: Add one figure per mechanism above. Prompts for generating them are in Working documents/image_prompts.md; a generated figure is an illustration and must be captioned as schematic, and the AI use belongs in Appendix A.]

3.1.4 Abstraction into design principles

Table 8 keeps the three columns strictly separate, as the method requires: what was observed, what the observation abstracts to, and what it might become technically. Only the third column contains anything that could be built, and nothing in it has been selected yet — that is the work of Section 3.2.

One principle, found three times. The jackrabbit and the elephant regulate by how much *area* is active, the bees by how many *units* have switched, and a bilayer shading element by how far it has *deployed*. All three modulate how many elements act rather than how strongly each one acts, and all three arrive at a graded response without a controller. That convergence from three unrelated organisms is the strongest abstraction this project has produced, and it is the one carried into every concept in Section 3.2.

3.1.5 Life’s Principles evident in the selected strategies

The principles below are the ones the *selected models* actually exhibit; Section 4 assesses how well our final concept applies them, which is a different question.

- **Be locally attuned and responsive** — present in every model that couples its response to the local load, and named by the checklist in the form our F7 requires: the signal, the antenna that detects it and the response must match (Baumeister et al. 2014).
- **Be resource efficient (material and energy)**, sub-principle *use low energy processes* — none of the selected mechanisms consumes energy; the load itself drives them.
- **Adapt to changing conditions**, sub-principles *incorporate diversity* and *maintain integrity through self-renewal* — *Encelia* and the pine cone for the first, the lichen for the second.
- **Use life-friendly chemistry** — *Cyphochilus* achieves with geometry what industry achieves with titanium dioxide. The lichen is the more careful case: its water repellency does involve a compound, but that compound works *with* a multi-scale geometry rather than instead of it, and nothing is released to do the job.

[TODO: Section 4 must *assess* these, not list them again. The rubric marks down a list.]

Table 8: From biological observation to transferable design principle. The third column lists possibilities, not decisions.

Biological observation	Abstracted principle	Possible technical interpretation
The snail's shell reflects strongly <i>and</i> an air space separates it from the hot substrate	Reflect first, then decouple whatever still gets through with a layer of air	Reflective outer skin over a ventilated cavity; structural white instead of a pigment
<i>Morpho</i> colour comes from layer geometry, not pigment, and the disordered structure stays diffuse	Separate what a surface looks like from what it does with heat	Structurally coloured or near-infrared-selective finish that keeps the existing façade colour
Cactus peaks shade the troughs; cooler trough air absorbs heat and rises out	Use folded geometry for self-shading, more emitting area and better airflow	Ribbed or profiled façade and paving relief
Sparse hairs reach through the still air layer clinging to the skin	Break the boundary layer so moving air reaches the surface	Millimetre-scale texture on façade and pavement
The jackrabbit varies blood flow to a thin radiator; the elephant opens patches	Regulate by varying the <i>thermal connection</i> between the store and a radiating surface, and place that radiator only where conditions favour loss	Pumpless density-driven liquid loop from the canyon face to a parapet-level radiator with a good sky view
Guard cells open and close the pore in response to heat and dryness	Build the control into the opening, so the escape route widens with the load	Load-driven vent in a ventilated cavity, opened by the heat itself rather than by a controller
Worker bees each switch at a different temperature	Spread many binary elements across a range of trigger points and the sum responds continuously	Any switching element built as many small units with a deliberate spread of switching temperatures
Pine cone scales bend because two bonded layers swell differently	Make moving parts from two layers that respond differently and let the environment actuate them	Bilayer shading element; hygroscopic or thermal actuation, no motor
The sea star charges on cold water sized by the previous hot exposure	Charge the store during the cool phase, sized by the load of the previous cycle	Night-charged thermal store, with an explicit reset condition across the five-day case
Lichen roughness at several scales sheds water while staying breathable	Use geometry rather than chemistry to shed water and stay clean	Multi-scale textured finish; no biocide in the film

3.2 Creating: Engineering Concept Development

3.2.1 Ideation

Ideation started from the morphological matrix rather than from free sketching, because the matrix guarantees that every idea is a path across our own functions instead of a shape we happened to like. [TODO: Add the Crazy 8s sheets: eight technically different transfers of Table 8, generated without evaluating. The three combinations below were read off the matrix afterwards and are what survived; the sheets are the evidence that the field was wider than three.]

Three combinations were read off Table 9 and checked for physical compatibility before any was developed further.

1. **Water façade.** Ribbed self-shading skin (F2, F3) + thermal mass carried in a liquid (F6) + a buoyancy-driven loop to a radiating surface with a good sky view (F4, F5). Self-regulation comes from the loop itself.
2. **Roughness spray.** Bright finish (F2) + boundary-layer texture for convection (F5) and nothing else. Self-regulation comes from the convective term.
3. **Switching coating.** Thermochromic finish that darkens in winter and brightens in summer (F2) + no removal path. **Dropped:** it acts only on what is absorbed, so it has no effect in the night window — which Section 2.1 identifies as the larger half of the problem and the half the incumbent already fails. It would have reproduced HA 11 with extra chemistry.

Combinations 1 and 2 were carried forward. They are separated by physics rather than by styling: the first *moves* heat to a place where it leaves more easily, the second lets moving air reach the surface where the heat already is. They also address different building stocks — the first needs pipework inside the construction and is a new-build system, the second is sprayed onto what already stands. The retrofit constraint in Table 2 is soft, which is what makes this pair admissible.

How biology was balanced against feasibility. Nothing in either concept reproduces a biological *form*. The elephant's ear contributes the *arrangement* — absorb in one place, release in another, and let the transport track the load — not a shape; the pipe network is conventional building services, and only its *pumplessness* is biomimetic. The cactus contributes a folded profile, which is manufactured as extruded or cast relief rather than as a plant silhouette. The elephant's hair contributes a length scale relative to the boundary layer, which becomes a sprayed aggregate, not a bristle. Where the biology offered something we could not build — vasodilation as an actively controlled valve — we kept the principle and dropped the actuation.

3.2.2 Morphological matrix

The method is function-based concept generation (Nagel et al. 2010): the rows of the matrix are functions, the columns are alternative principles that serve them, and a concept is a path across the rows. The rows are chosen, not inherited. Of our eight functions, a row belongs in the matrix only where alternative principles would lead to a meaningfully different concept. On that criterion F1 is excluded as the main function, F8 is a requirement and an engineering check rather than a source of concept variants, and F7 is included only if the mounting really changes the architecture. That leaves four rows.

The two paths share no cell. That is the test of whether a pair is genuinely distinct, and it is why the switching coating was the one dropped: its path would have overlapped B in three of four rows.

Table 9: Morphological matrix. Rows are the four functions for which alternative principles produce genuinely different concepts; columns are the technical interpretations abstracted in Table 8. **A** marks the path taken by the water façade, **B** the path taken by the roughness spray.

Function	Option 1	Option 2	Option 3
F2, F3 Reduce heat input	Structural white or near-infrared-selective finish (B)	Ribbed, self-shading geometry (A)	Deployable shading element
F4, F5 Move heat away	Broadband emitter with a good sky view (A)	Boundary-layer texture for convection (B)	Density-driven liquid loop to a parapet radiator (A)
F6 Store and buffer	Thermal mass in the build-up	Thermal mass carried in a liquid (A)	No added storage; decoupling only (B)
F7 Couple response to load	Switching material, many units, spread thresholds	Buoyancy: the driving head grows with the temperature difference (A)	Convection: the flux grows with the surface-air difference (B)

3.2.3 Concept A — the water façade

What it is. A new-build façade whose outer skin is a ribbed, light-coloured panel and whose build-up carries a closed water loop. The ribs shade the troughs between them, as a cactus does, so part of the surface is never in direct sun. Behind the panel, wide-bore risers absorb what the surface does take up. Water warmed in the risers becomes less dense and rises on its own; cooled water in the return line sinks. At the top the loop opens into a thin, wide plate at roof level, where the water is spread over the largest area the roof allows and gives its heat to the night sky and to the air above the building. Nothing is pumped, nothing is controlled and no water is consumed — the loop is closed.

Functional chain. F2 and F3 at the ribbed surface reduce what is absorbed; F6 stores what is absorbed in the water rather than in the masonry; F5 and F4 release it at the roof plate; and F7 — the coupling the design question asks for — is not an added device but the physics of the loop itself. The warmer the façade, the larger the density difference between riser and return, the larger the driving head, and the faster the circulation. The response scales with the load without a sensor.

Where the biology enters, and where it does not. Table 10 separates the two, as the method requires.

Materials and parameters. [TODO: Fix the panel material and its α_{sol} and ε ; the pipe material and its corrosion and freeze behaviour; the loop volume; and the roof plate construction. The bore follows from Engineering check 1 below and is not free.]

3.2.4 Concept B — the roughness spray

What it is. A sprayable coating carrying a millimetre-scale aggregate, applied to an existing bright façade. The particles stand proud of the surface far enough to reach through the layer of still air that clings to any smooth wall, so that moving canyon air touches the surface instead of gliding over an insulating film. It adds no moving parts, no water and no construction; it is applied by ordinary trades, is close to invisible from the street, and is renewed by spraying again.

Table 10: Concept A: which parts are transferred from biology and which are ordinary engineering.

Component	Origin	What was transferred
Ribbed outer panel	Cactus rib (AskNature 2026e)	Folded geometry that shades itself and enlarges the emitting area. The profile is extruded, not plant-shaped
Riser and return	Elephant ear, jackrabbit ear (AskNature 2026i; AskNature 2026h)	The <i>arrangement</i> : absorb in one place, release in another, and let the transport rate follow the load. The pipework itself is conventional
Buoyancy drive	—	Ordinary thermosiphon physics. The biology suggested pumplessness; it did not supply the mechanism
Roof plate	Elephant ear	Maximum thin area at the place with the best view of the sink
Charging behaviour	Sea star (AskNature 2026c)	Charge during the cool phase, sized by the previous cycle; the reset condition of Section 2.6 comes from here

Functional chain. F2 at the bright finish, F5 at the texture. There is no F6 and no F4 beyond what the wall already does: B deliberately buys one function well rather than four badly. F7 comes from the convective term itself — the flux is proportional to the surface-to-air difference, so the response rises with the load, linearly rather than steeply.

Where the biology enters. The elephant’s sparse hairs sit *below* the density at which hair insulates, so each acts as a fin through the boundary layer rather than as part of a blanket; the reported effect is an additional heat loss of up to 23 % (University of Notre Dame 2021). The transferred quantity is a *length scale relative to the boundary layer*, not a bristle. The lichen contributes the second requirement: multi-scale roughness that sheds water and stays clean without a biocide (Biomimicry 3.8 and HOK 2013, pp. 41–44), which is what keeps the texture from failing criterion 6. That this is manufacturable rather than speculative is established by prior art: *StoColor Lotusan*, a micro-structured superhydrophobic façade paint modelled on the lotus leaf, has been sold since 1999 and is stated by its manufacturer to resist algae and fungal growth *without biocide film protection* (Sto SE & Co. KGaA 1999).

The obvious objection. Roughness can also *thicken* the still layer instead of piercing it, if the elements are too short or too dense. This is the concept’s decisive assumption and it is question 2 below. **[TODO: Fix the particle size, the areal density and the binder, and state the expected h_c before and after.]**

3.2.5 Engineering Canvas and critical questions

A critical question names a quantity, a threshold and a scale. Three were derived from the canvas and decide between the concepts rather than merely describing them.

1. **Does the loop drive itself?** Is the buoyancy head over the available building height enough to move the required flux through a pipe bore that still fits the build-up thickness allowed in Table 2? Checked below.
2. **Does the texture pierce the boundary layer or thicken it?** At the low wind speeds of a street canyon, does millimetre roughness raise the convective coefficient, or does it trap a deeper stagnant film and make things worse? This single question decides whether B

Table 11: Engineering Canvas for both concepts.

Field	A — water façade	B — roughness spray
Function	F2, F3, F6, F5, F4, F7	F2, F5, F7
Biological mechanism	Ear-based transport; cactus rib; sea-star charging	Sparse hair as a boundary-layer fin; lichen roughness
Abstracted principle	Move heat to where release is easy; let the load drive the transport	Break the still layer so moving air reaches the surface
Technical principle	Buoyancy-driven closed water loop to a roof radiator	Sprayed millimetre aggregate on a bright finish
Architecture	Ribbed panel → riser → roof plate → return	Substrate → primer → textured top coat
Inputs / outputs	In: solar gain. Out: heat to sky and air. No mass crosses the boundary	In: solar gain. Out: heat to canyon air
Requirements	Criteria 1–6; build-up thickness and dead load	Criteria 1, 2, 6; installability
Assumptions	The buoyancy head is enough; the roof area is enough; the loop survives winter	The texture pierces rather than thickens the boundary layer
Uncertainties	Freezing; roof area; flow distribution across risers	Soiling; abrasion; whether the gain survives at low wind
Next check	Checks 1 and 2 below	[TODO: wind-tunnel or plate comparison]

works at all. **[TODO: Check by measurement: two identical plates, one sprayed, logged over a hot afternoon.]**

3. **Where does the heat actually go?** Is the roof area large enough to release what the façade collects? Checked below, and the answer changed the design.

3.2.6 Engineering check 1: does the loop drive itself?

A thermosiphon circulates because the fluid in the warm leg is less dense than in the cool leg. The available driving pressure over a loop of height h is

$$\Delta p = \rho g \beta \Delta T h, \quad (1)$$

where ρ is the density of water, g the acceleration of gravity, β its volumetric thermal expansion coefficient, ΔT the temperature difference between riser and return, and h the height between the collecting and releasing sections.

Result. Equation (1) gives

$$\Delta p = 995 \times 9.81 \times 3.0 \times 10^{-4} \times 10 \times 16 \approx 470 \text{ Pa}.$$

The flux to be carried is $\dot{Q} = \dot{q}A = 100 \text{ W/m}^2 \times 288 \text{ m}^2 \approx 29 \text{ kW}$, which at $\Delta T = 10 \text{ K}$ requires a mass flow of $\dot{m} = \dot{Q}/(c_p \Delta T) \approx 0.7 \text{ kg/s}$. Distributing that over a loop of roughly 40 m developed length leaves about 12 Pa/m of allowable friction. A bore of 50 mm gives roughly 34 Pa/m and fails; a bore of about 80 mm gives roughly 4 Pa/m and passes with margin for bends and the plate.

What this does and does not prove. It establishes that the concept does not need a pump, which is what the energy constraint demands, and it fixes a design parameter that was previously free: the risers must be of the order of 80 mm, not the 15 mm of ordinary heating pipework. That is an architectural consequence — it sets the build-up thickness against the soft constraint in

Table 12: Assumptions for Engineering check 1.

Quantity	Value used	Justification
h	16 m	Five storeys plus roof structure, the reference site of Section 2.2
ΔT	10 K	Assumed riser-to-return difference under full load. [TODO: confirm against the transient model]
β	$3.0 \times 10^{-4}/\text{K}$	Water at about 30 °C
ρ	995 kg/m ³	Water at about 30 °C
$\dot{q}$	100 W/m ²	Absorbed flux at $\alpha_{\text{sol}} = 0.8$ under a façade irradiance of about 500 W/m ² . [TODO: take the orientation-specific irradiation from (Swiss Federal Office of Energy 2026) and from the transient model, and report the value used]
A	288 m ²	18 m wide $\times$ 16 m high, one façade of the reference block
c_p	4180 J/(kg K)	Water

Table 2, and it is the reason Concept A is a new-build system. It does *not* prove that the flow distributes evenly across parallel risers, and it says nothing about winter. **[TODO: Repeat for $\Delta T = 3 \text{ K}$, the weakest driving condition, and report whether the loop stalls.]**

3.2.7 Engineering check 2: where does the heat go?

The loop can only release what its releasing surface can pass. For a surface close to ambient the net release is of the order of 80 W/m² to 100 W/m² at night, combining long-wave emission to a clear sky with free convection (Raman et al. 2014; Zhai et al. 2017). The required area follows directly:

$$A_{\text{release}} = \frac{\dot{Q}}{q''} = \frac{29 \text{ kW}}{100 \text{ W/m}^2} \approx 290 \text{ m}^2. \quad (2)$$

Result, and what it changed. The roof of the reference block is about $18 \times 14 \text{ m} = 252 \text{ m}^2$. The roof is therefore *too small* to release what one façade collects, before any allowance for plant, access or photovoltaics. This is the most useful result in this section, because it moved the problem: the binding limit on Concept A is not the driving pressure — check 1 passes comfortably — but the area available to give the heat away. Section 3.3 carries this into the selection, and it is the weakness Concept V2 answers. **[TODO: Recompute q'' for our own sky and emittance assumptions and report a range; and state how much area photovoltaics would claim.]**

3.3 Concept Evaluation & Iteration

The two concepts are compared against the six criteria of Section 2.6, using the two engineering checks as evidence rather than as decoration. Where a rating rests on an estimate rather than a computed number, the evidence column says so.

The last three rows carry the questions a criterion table cannot answer: whether the concept can be built, whether it genuinely transfers the biology or merely resembles it, and how well supported the ratings above are. The third is the one that prevents a confident-looking table from concealing guesswork, and it is where B scores worst: its decisive assumption has not been tested.

Table 13: Decision matrix. Ratings 1–5. The last three rows are not criteria and are deliberately kept apart from them.

Criterion	Wt	A	B	Evidence used
1 Cooling slope	[TODO:]	5	3	A: the driving head grows with ΔT , so the transport rate itself follows the load (check 1). B: the convective flux is linear in $T_s - T_a$ — positive, but not steepening
2 Reflectance	[TODO:]	4	4	Both keep a bright finish; neither has an advantage
3 Surface cooling	[TODO:]	5	3	A removes absorbed heat from the wall instead of only reflecting it. B raises the convective term only
4 Street-level comfort	[TODO:]	5	2	A releases at roof level, away from head height. B releases <i>into the canyon</i> , where the people are — a real disadvantage that the criterion exposes
5 Night release	[TODO:]	4	2	A: the loop runs after sunset and the roof plate sees a clear sky. B: no store, so little is left to release
6 Staying power	[TODO:]	2	5	A: a buried water loop cannot be renewed without opening the façade. B: sprayed on, sprayed again — the strongest single argument for B
Feasibility	[TODO:]	2	5	A needs new construction, 80 mm bores and freeze protection. B needs a spray gun
Biomimetic transfer	[TODO:]	5	4	A transfers an arrangement and a control principle; B transfers a length scale
Evidence and uncertainty	[TODO:]	3	1	A has two checks with numbers. B rests on an untested assumption about the boundary layer
Weighted total		[TODO:]	[TODO:]	[TODO: fill once the weights are agreed]

What the comparison shows. A leads on five of six criteria and B on one — but the one it leads on is durability, which is exactly where A is weakest, and B is far ahead on feasibility. Two concepts whose strengths are complementary rather than overlapping are an argument for a hybrid, not for a winner.

3.3.1 Review and feedback

[**TODO: World café peer feedback, instructor feedback, and AI used as a critical reviewer. For every comment record the decision — accept, reject or investigate — and the justification. A documented rejection is worth more than a list of accepted comments. This belongs in Appendix A as well.**]

3.3.2 Selection and Final Concept V2

Selected: a hybrid, A with B applied to it. Neither concept is carried forward alone. Engineering check 2 showed that the roof cannot release what one façade collects — 290 m^2 required against 252 m^2 available — so Concept A as first drawn is limited not by its physics but by the area of its one releasing surface. Concept B raises the release rate of *any* surface it is sprayed on, at negligible cost and with no construction.

What V2 changes. The roof plate stops being the only radiator. In V2 the ribbed façade panels are part of the return path, so the façade itself releases at night what it collected by day, and the roughness spray is applied to both the façade panels and the roof plate to raise the convective coefficient on each. The releasing area is then the façade plus the roof rather than the roof alone, which is what closes the gap check 2 opened. A second consequence follows without being designed for: the spray is renewable, so the part of the system that wears is also the part that can be replaced — which repairs A’s weakest criterion.

Table 14: Final output of the creating and evaluating phase: Concept V1a/b → engineering review → evidence → Concept V2.

What changed	Why it changed	Evidence that informed the change
The façade became a radiator as well as a collector	The single roof plate could not pass the collected flux	Engineering check 2: 290 m^2 required, 252 m^2 available
The roughness spray was applied to A rather than kept as a rival	It raises the release rate of every surface it touches, and it is renewable	Decision matrix: B leads criterion 6, the one criterion on which A scores 2
The switching coating was dropped before development	It has no effect in the night window	Section 2.1: -0.1 K at night is the incumbent’s failure, not something to reproduce

3.3.3 Remaining uncertainties

Four are known and none is closed. **Does the texture pierce or thicken the boundary layer?** B’s decisive assumption is untested, and V2 now depends on it in two places rather than one. **Does the loop survive a Swiss winter?** The climate constraint requires operation to about -10°C ; freeze protection without auxiliary energy is not designed. **Does the loop still drive itself at small ΔT ?** Check 1 was run at 10 K ; at 3 K the driving head falls to about 140 Pa and the margin may vanish. **Does millimetre texture survive twenty years?** It sits on the outside of the building, and criterion 6 asks for 90 % retention after three. [**TODO:**

Rank these by how much each would change the design if it failed, and say which one you would test first and how.]

4 Sustainability & Life's Principles

4.1 What this section assesses, and what it does not claim

Section 2.6 named a Life's Principle beside each performance criterion. Naming them is not applying them, and the difference is the whole content of this section: each principle below is stated as a *claim*, followed by the evidence for it and by the point at which the concept falls short of it. A principle with no shortfall named is a principle that has not been assessed.

Two limits are stated at the outset rather than buried. First, the concept assessed here is Concept V2 of Section 3.3 — a ribbed new-build façade carrying a closed, pumpless water loop, with a sprayed millimetre texture on the façade panels and the roof plate. It is a described concept, not a dimensioned design, and the assessment can be no more specific than that. Second, no life cycle assessment was carried out. Where embodied emissions are discussed they are discussed as a *criterion with a decision rule*, not as a computed result, and the words “sustainable” and “resource efficient” are avoided as descriptions of the concept because on the present evidence neither could be defended.

4.2 Life's Principles applied to the concept

4.3 Eco-efficient or eco-effective?

Cradle to Cradle draws a distinction that this project has to face directly (Braungart and McDonough 2002). An *eco-efficient* design is less bad: it does the same thing while emitting less. An *eco-effective* design is actively good: its outputs are inputs somewhere else. The standard illustration is a brake pad — one version emits fewer particles but still emits thousands of tonnes, the other is made of a material that can safely re-enter a biological cycle. Nature, the argument runs, is not efficient at all: a cherry tree produces blossom wastefully, and is nonetheless effective because every part of that waste is a nutrient.

Measured against that distinction, Concept V2 is **eco-efficient and no more**. A façade that absorbs less, moves what it does absorb to a better surface and gives it away without energy is a less bad façade; it produces no nutrient for anything.

One thing is worth stating precisely, because it is easy to overclaim. The loop is *closed*: no water is consumed, none evaporates, and nothing is drawn from the municipal supply. Against an evaporative façade — the obvious alternative, the direction an earlier version of this scoping carried, and one already built as *HydroCanopy* (AskNature 2026f) — that is a genuine advantage under the water constraint of Table 2, and it is the honest way to state it: V2 avoids a resource cost rather than creating a resource benefit. Avoiding a cost is efficiency. It is not effectiveness.

An eco-effective version of this concept would have to make its output the input of something else. Two routes exist and both were placed outside the boundary in Section 2.3: recovering the collected heat for domestic hot water, which the boundary excludes as “reuse of the stored energy” because it turns the project into a building-services design; and harvesting the rainwater that falls on the same roof plate, which would reconnect the discarded stormwater the parallel-system comparison identified. Neither is pursued here, and the honest summary is that the project remains at the efficient end by a boundary decision rather than by a physical limit. That is the first thing a follow-on project should revisit.

4.4 The material perspective: which layer belongs in which cycle

Cradle to Cradle divides materials into *biological nutrients*, which are consumed and must return safely to the environment, and *technical nutrients*, which are used without being consumed and

Table 15: Life's Principles assessed against the concept. The last column is the part that matters: a principle claimed without a shortfall has not been assessed.

Principle and sub-principle claimed	How the concept applies it	Where it falls short
Be locally attuned and responsive <i>the signal, the antenna and the response match</i>	F7 is this principle expressed as a function: the surface temperature is the signal, the physics of the removal path is the antenna, and the cooling power is the response. The coupling is not a device but the physics of the loop: the warmer the façade, the larger the density difference driving the circulation. The sinks it uses — the night sky over the block and the canyon air — are local and freely available, and the diurnal cycle it works against is the site's own.	The response is attuned to <i>temperature</i> , not to the quantity that actually matters to people, which is radiant load at eye level. The concept optimises what it can sense, not what it is for. [TODO: Revisit once V2 is fixed.]
Be resource efficient <i>use low energy processes and one design meets more than one need</i>	The cooling function consumes no auxiliary energy: Engineering check 1 shows the loop drives itself, so the constraint is met by physics rather than by restraint. No water is consumed either — the loop is closed. V2 also earns <i>one design meets more than one need</i> : after check 2 the façade panels collect by day and release by night, so one surface carries two functions instead of two surfaces carrying one each.	Nothing has yet been counted, and the water loop makes the count worse rather than better: several hundred metres of wide-bore pipe and a roof plate are a substantial addition. “Resource efficient” is a claim we cannot make until those embodied emissions are set against the cooling delivered over the service life.
Adapt to changing conditions <i>adapt to temporal and spatial changes, maintain integrity through self-renewal</i>	The design case is a five-day heat wave with a reset condition, not one hot afternoon. <i>Maintain integrity through self-renewal</i> is met literally by the sprayed layer: the part of V2 that wears is the part that can be renewed without opening the construction, which is what repairs the weakest criterion of Concept A.	The seasonal half of the principle is weakly served. A surface optimised for July is a surface that also faces January, and the winter trade-off is currently a constraint we accept rather than a behaviour we design.
Use life-friendly chemistry <i>safe subset, harmless breakdown products</i>	The decision that self-cleaning must be physical rather than chemical follows directly from this principle: anything abraded from the sprayed texture lands in the street canyon the project exists to improve.	V2 introduced a new conflict with this principle. The loop must survive frost to about -10°C , and the ordinary answer is glycol — a substance in a pipe that runs the full height of an occupied façade and whose failure mode is a leak into the wall and the ground. The principle is otherwise satisfied by <i>exclusion</i> only: no substance we intend to use has yet been checked in a toxicity database. See Section 4.7.
Integrate development with growth <i>built to shape</i>	A modular element sized to a façade grid produces no offcuts. On-site material waste in construction is put at 10 % to 15 % by the Ellen MacArthur Foundation (Ellen MacArthur Foundation 2015), so this is a measurable quantity rather than a gesture.	Claimed narrowly and on purpose. The sub-principle <i>creates more opportunities for life</i> is not claimed: a permanently damp textured surface invites exactly the algal growth KPI 6 penalises.

are recovered (Braungart and McDonough 2002). The rule that assigns a material to one or the other is about wear:

Anything that is consumed — worn out — has to come from the biological cycle.
Anything that is merely used, without wear, can come from the technical nutrient cycle.

Applied layer by layer, this decides the material strategy of the build-up before any material has been chosen, and it produces one uncomfortable result.

Table 16: The wear rule applied to the build-up.

Layer	Worn or used?	Consequence
Sprayed texture layer	Worn — UV, driving rain, freeze–thaw, abrasion	Must be biologically benign in breakdown: it is abraded straight into the street canyon below. It is also the layer designed to be renewed, so it is consumed repeatedly over the service life
Ribbed panel and its finish	Worn slowly, at the surface	Same requirement, at a slower rate
Pipework, roof plate, sub-structure, fixings	Used , not consumed	Technical cycle: recoverable and worth specifying as separable single materials. The bore established in Engineering check 1 makes this the largest material mass in the build-up, so its recoverability matters more than for a thin cavity
Heat transfer fluid	Used , but mobile	Neither cycle cleanly. It stays in the loop for decades, then must be drained and taken back — and if frost protection requires an additive, it is the one component that can leave the system by accident
Masonry behind	Untouched by the retrofit	Outside the product

The uncomfortable result. The layer we would most readily engineer with chemistry — the sprayed texture, where the convective performance lives — is precisely the layer that must be biologically benign, because it is the one designed to wear away, and it wears into the place we are trying to improve. That inverts the usual order of a coating specification, in which durability and optical performance are chosen first and the environmental profile is checked afterwards.

4.5 Circular design principles: an honest status

Items 6 and 7 are the honest weaknesses here, and they share a cause: both need somebody to own the material after its first life, which needs a business model — rental, service or deposit — rather than a sale. In a sales model ownership transfers, so the value created by durable, recoverable design accrues to the recycler, the user and the maintenance trade rather than to whoever paid for it. That split incentive is named as the principal obstacle to circular construction (Ellen MacArthur Foundation 2015), and this project has it in an acute form: the owner pays, the tenant gains comfort, the street gains the cooling, and nobody owns the outcome.

4.6 Environmental benefits, limitations and trade-offs

Summer against winter. A high solar reflectance reduces summer heat gain and also reduces useful passive solar gains in winter. The constraint in Table 2 requires that heating demand

Table 17: The eight cycle design principles, with the status of each. Three can be answered today, three depend on the final concept, and two lie outside the project boundary.

	Principle	Status	Evidence or reason
1	No ecotoxic chemicals	Answered	Ruled out by constraint; the self-cleaning function must be physical (Section 4.7)
2	Recyclable materials	Partly answered	The pipework and roof plate are single-material and recoverable; the panel-plus-texture composite is not. [TODO: name the panel and binder materials]
3	Easy to dismantle, modular	Partly answered	Panels and pipework are demountable in principle; the sprayed texture is bonded to the panel and cannot be separated from it
4	Durability	Answered	KPI 6: $\geq 90\%$ of the cooling function retained after three years; service life target 20 years
5	Maintainability and reparability	Answered for the wearing layer	The texture is renewed by spraying again, without opening the construction. The loop behind it is not repairable without doing so
6	Cascade use of raw materials	Not addressed	Nothing in the concept anticipates a second life for its materials
7	Planned return logistics	Out of boundary	Requires a business model; Table 4 places this outside the project
8	Efficient use of energy	Answered	Hard constraint: no auxiliary energy for the cooling function

must not appreciably rise. **[TODO: Estimate both halves for the Swiss climate and give the net annual balance. Until that number exists this is a known unquantified risk, not a solved trade-off.]**

Local against global. The concept moves heat out of the street canyon. Whether it *reduces* the total energy in the system or merely *redistributes* it is a different question, and the honest answer differs by mechanism, and V2 uses both. The roof plate radiates to the sky, which sends energy out of the system and is a genuine removal. The sprayed texture works by convection, which relocates heat into the canyon air rather than removing it — and on the façade that air is at head height. Criterion 4 already penalises this: it is why Concept B alone scored 2 there. **[TODO: State which of these the final concept relies on, and do not claim a reduction where the mechanism only redistributes.]**

Frost protection against life-friendly chemistry. This is the trade-off V2 created and it has no comfortable answer. The loop must survive to about -10°C without auxiliary energy. Glycol is the standard solution and would put a mobile additive inside a pipe running the full height of an occupied façade, where the failure mode is a leak into the wall or the ground. Draining for winter avoids the substance but also stops the system working in the shoulder seasons and adds an operation somebody must remember. A third route — sizing and routing the loop so that it is self-draining by gravity — keeps the chemistry clean and is the one worth checking first, because it is the only option that does not trade a principle for a season. **[TODO: Establish which of the three is feasible, and if an additive is unavoidable, name it and check it as Section 4.7 requires.]**

Materials. [TODO: Panel and binder chemistry, the aggregate in the spray, pipe material and its corrosion behaviour, separability at end of life, and the embodied greenhouse gas of the added build-up — now dominated by the pipework — against the constraint in Table 2.]

4.7 Circular does not mean sustainable

Keeping a material in circulation is not by itself an environmental benefit. The standard counter-example is an additive that is harmless in its first application, in a laptop housing, and hazardous in its second after recycling into a children's toy: the substance did not change, the exposure pathway did.

Our project has the same failure mode one step earlier, and it is the reason the life-friendly-chemistry principle is not decorative here. A biocide or a stabiliser that is entirely acceptable *in* a façade coating becomes a different question once it leaves that coating — and in V2 the sprayed texture is not merely worn, it is *designed* to be consumed and renewed, so everything in it reaches the street eventually and by intention rather than by failure. What it wears into is the street canyon directly below — the same pedestrian-level space Section 1 names as the place the whole project exists to improve. The concept would be depositing its own emissions exactly where its beneficiaries stand.

Two consequences follow. The self-cleaning function must be structural rather than chemical — which the lichen strategy of Section 3.1 shows is physically possible and the commercial lotus-effect coating cited in Section 2.1 cites as prior art shows is manufacturable. And any substance that does remain in the specification must be checked rather than assumed.

[TODO: Do this check and report it: name the substance, the database, and the result. The REACH registry at the European Chemicals Agency (European Chemicals Agency 2026) is the primary source; PubChem and the EPA DSSTox database are alternatives. One short paragraph naming a substance, a database and a finding is worth more here than a page of general language.]

4.8 Did sustainability considerations change the design?

The template asks this directly, and it is the question a sustainability section most easily fails, because the honest answer is often no. Here it is yes, in three documented instances, each traceable to a decision recorded in Table B.1 in Appendix B.

1. **The reflectance target was lowered from what is technically possible.** Ultra-white coatings reach a solar reflectance of about 0.98 (Li et al. 2021), and an earlier version of this project treated that as the goal. KPI 2 now targets ≥ 0.80 , the city's own threshold, because the principle of optimising rather than maximising makes the further reflectance a cost in glare, winter gains and photovoltaic yield rather than a benefit. The performance target was changed by a sustainability argument.
2. **The self-cleaning function was constrained to a physical mechanism.** The obvious solution to soiling is a biocide in the film. The leaching argument of Section 4.7 removed that option before any concept was developed, which is why the lichen strategy entered the model table at all.
3. **The wearing layer was chosen to be the renewable one.** In Concept A the part that degrades sits behind the panel and cannot be reached; in V2 it is the sprayed texture on the outside. That was selected on a durability and maintainability argument (Table 17, items 4 and 5) rather than on cooling performance, and it is why the hybrid was preferred to A alone.
4. **Cost and embodied emissions were reclassified.** They were originally an eighth KPI. Recognising that a KPI must follow from a function moved them into the constraints of

Table 2 — where a violation disqualifies a concept rather than merely scoring against it. The change made the requirement stricter, not weaker.

4.9 Limits of this assessment

No life cycle assessment was performed, so no statement in this report should be read as a quantified environmental balance. The embodied emissions of the added build-up are a constraint with a decision rule and not a measured figure. The assessment of Life's Principles above is made against a concept direction rather than a finished design, and **[TODO: must be repeated against Concept V2 once Section 3.3 selects it]**. Finally, two of the eight circular design principles — cascade use and return logistics — are not addressed at all, and the reason is a boundary decision rather than an analysis: they require a business model, and this project does not design one.

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

[TODO: Be concrete and honest. Which idea came specifically from the biological model, and which would a conventional engineering approach have produced anyway? A well-argued “this part was conventional” is worth more than an overstated claim.]

5.2 Strengths and weaknesses of the final concept

[TODO:]

5.3 Assessment against the challenge and the KPIs

[TODO: Go back through the six criteria of Section 2.6 and state for each one whether it was met, missed, or could not be assessed with the available evidence. Add the three Life's Principles used as criteria in Section 2.6, and say for each whether the concept meets it — Section 4 already names where it does not.]

5.4 Innovative character and limits of the evidence

[TODO: State clearly what has been demonstrated (calculation, measurement, literature) and what remains an assumption.]

6 Glossary

Short-wave radiation The radiation the sun delivers, roughly 300 nm to 2500 nm (ultraviolet, visible and near-infrared). It is what a surface can *reflect*; the fraction reflected is the solar reflectance α_{sol} of criterion 2. Function F2 acts on it.

Long-wave radiation The thermal radiation a surface at ordinary temperatures emits by itself, roughly 4 μm to 50 μm . It is what a surface can *give away*, including at night and to a clear sky, and it is governed by the emittance ε rather than by α_{sol} . Function F4 acts on it. The two are separate properties: a surface can be bright in the short wave and still a poor emitter in the long wave, which is why criterion 2 and criterion 5 are reported apart.

Atmospheric window The band near 8 μm to 13 μm in which a clear sky is largely transparent, so long-wave radiation emitted there escapes to space rather than being returned. It is what makes night-time radiative cooling below air temperature possible.

- Albedo** The share of incoming short-wave radiation a surface reflects. The city classes ≤ 0.30 as low and ≥ 0.80 as high (Stadt Zürich 2020a, p. 128).
- PET** Physiologically equivalent temperature: the temperature a person would experience given air temperature, radiation, humidity and wind. It is the metric the client uses to judge a measure at street level, and it is a different quantity from surface temperature.
- UHI** Urban heat island: the persistent temperature difference between a city and its rural surroundings, caused by urban surface properties and geometry (Oke 1982). In Zurich it reaches about 7 K at night (Stadt Zürich 2021).
- Solution-neutral** A statement of *what* must happen that names no means of achieving it. Design questions and functions in this report are written this way so the biology can propose the means.

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Appendix A – AI Use Statement

Relevant AI-supported activities are documented below. Not every prompt is listed; the focus is on how AI contributed to the work and how important outputs were verified. AI output is not used as evidence for factual claims.

Table A.1: AI use statement.

AI tool	Used for	How the output influenced the project	How important output was verified
AskNature (search and strategy pages)	Discovering biological strategies for functions F1–F8, navigated by taxonomy branch rather than by organism	Produced the candidate list in Table A.2, including four models that were subsequently de-selected	Every mechanism claim carried into the report is referenced. The pages could not be opened programmatically, so titles and URLs were taken from search results; the open item in Section 3.1 records that each must be confirmed and, where a primary paper exists, replaced by it

AI tool	Used for	How the output influenced the project	How important output was verified
Large language model assistant	Structuring the scoping, distilling the course material into working notes, drafting report text, building the transient wall model of Section 3.2, and generating prompts for figures	Draft text and table structures throughout Sections 1–4; the search-path table; the transient model and its figures	No factual claim entered the report without a named source. Benchmark values were checked against the Fachplanung. Two errors in the AI-built wall model were found and corrected: an air-cavity node made the explicit scheme unstable, and the synthetic weather drive had an inverted diurnal phase, which showed up as a peak surface temperature exactly equal to the air temperature
AI as critical reviewer	[TODO: Section 3.3 asks for this. Give the reviewer the concept sketch, the Engineering Canvas, the functions and KPIs, and the assumptions]	[TODO: Record each comment and the team's decision: accept, reject or investigate]	[TODO: A documented <i>reject</i> with a reason is worth more here than a list of accepted comments, because it shows judgement]
[TODO: Further tools used by the team]	[TODO:]	[TODO:]	[TODO:]

On the limits of this statement. Nothing in this report is cited to an AI tool. Where an AI-assisted search surfaced a candidate, the candidate was carried forward only with a source of its own, and the two remaining places where that chain is incomplete are marked as open items rather than left implicit: the AskNature pages listed above, and the 23 % figure for elephant hair, which traces to a university blog rather than to a study (University of Notre Dame [2021](#)).

Appendix B – Optional Supporting Material

B.1 Biological models found, and the selection decision

Referenced from Section 3.1. The criterion applied throughout: a model must be transferable to a *passive surface* that survives twenty years of the Swiss annual cycle.

Table A.2: Biological models found, and the selection or de-selection decision. The criterion applied throughout: a model must be transferable to a *passive surface* that survives twenty years of the Swiss annual cycle.

Organism / model	Function	Mechanism	Decision
<i>Prevention — reduce what is absorbed</i>			
Desert snail <i>Sphincterochila boissieri</i>	F2, F6	Shell reflects $\approx 90\%$ of visible and $\approx 95\%$ of near-infrared light; in extreme heat the aperture is sealed with a calcified plate and the animal withdraws, leaving an insulating air space between body and hot ground (AskNature 2026k)	Selected. Passive, mineral, non-metabolic — the closest analogue to an envelope we found
<i>Morpho</i> butterfly	F2	Transparent chitin–air layers a few hundred nanometres apart diffract and recombine light; colour comes from geometry, and the disordered structure keeps the reflection diffuse rather than specular (AskNature 2026d)	Selected, with a stated limitation. It decouples appearance from absorptance, which is what the townscape constraint requires — but the <i>Morpho</i> demonstrates structural <i>colour</i> , not near-infrared heat rejection. That transfer needs separate technical verification; the organism does not establish it
White beetle <i>Cyphochilus</i>	F2	Brilliant whiteness from a disordered chitin network only $7 \pm 1.5 \mu\text{m}$ thick, without pigment (Vukusic et al. 2007)	Selected as the achromatic case of the same principle
Saharan silver ant <i>Cataglyphis bombycina</i>	F2, F4	Triangular hairs scatter visible and near-infrared light and simultaneously raise emissivity in the mid-infrared (Shi et al. 2015)	Selected. The only model that does F2 and F4 in one structure
Cactus ribs	F3, F5, F4	Peaks shade the troughs; the cooler trough air absorbs heat and rises out, while the folds enlarge the emitting area and disturb the airflow around the body (AskNature 2026e)	Selected. Geometry only — no material, no water, nothing to bleach
Tree bark	F2, F4	Reflects part of the incoming radiation while emitting efficiently; rough layering shades itself and traps insulating air (AskNature 2026b)	Selected. A dead outer layer protecting a living structure — the same role our build-up has
<i>Removal — get heat out again</i>			

Organism / model	Function	Mechanism	Decision
Elephant ears	F4, F5	Warm blood is carried from the core into large thin ears; the surface releases it, and flapping raises the airflow (AskNature 2026i)	Selected. Separates the place of absorption from the place of release
Elephant skin hairs	F5	At this density the hairs do not insulate — it lies well below the crossover at which insulation becomes dominant — so each acts as a small heat-transfer <i>fin</i> , reaching through the still boundary layer that clings to the skin so that moving air reaches the surface. Reported additional heat loss up to 23% (University of Notre Dame 2021)	Selected. Directly relevant at the low wind speeds of a canyon
Jackrabbit ears	F7, F4	Vasodilation varies the blood flow to a thin radiating surface; at $\approx 30^\circ\text{C}$ air temperature the ears shed all excess heat, and the strategy exists to avoid spending water (AskNature 2026h)	Selected. The water-free route to a load-coupled response
Elephant skin hot spots	F7	Vasodilation in scattered patches rather than uniformly; the number of open “thermal windows” varies with the load (AskNature 2026g)	Selected. Regulation by active <i>area</i> rather than by material
<i>Self-regulation and supporting functions</i>			
Leaf stomata	F7	Environmental and chemical signals change the solute concentration in the two guard cells; osmosis moves water accordingly, turgor changes, and the cells bend open or shut the pore. The regulation is built into the opening itself (Biomimicry 3.8 and HOK 2013, pp. 62–65)	Selected for the valve, not for the transpiration. The clearest natural instance of an aperture whose width follows the load; the evaporative half is out of scope (Table 4)
Sea star	F6, F7	Takes up more cold water at high tide after a hot low tide, so the previous load sets the charge; fails once the sea is no longer colder than the air (AskNature 2026c)	Selected for the charging rule. Charge the store during the cool phase, sized by the previous cycle — including its own failure mode
Honeybee fanning thresholds	F7	Each worker begins fanning at a slightly different temperature, and the spread is genetic; the number of active bees rises with the load, so binary units produce a graded colony response (AskNature 2026l)	Selected. Resolves whether a threshold can satisfy F7
Pine cone scales	F3, F7	Two bonded layers swell differently with moisture: taking up moisture makes one expand more, so the scale bends and closes, and as it dries it returns towards the open state. The movement is reversible and the cone is dead tissue, moved entirely by the weather (AskNature 2026j)	Selected. A passive actuator with no energy input

Organism / model	Function	Mechanism	Decision
Lichen <i>Lecanora conizaeoides</i>	F8	Roughness at several scales keeps droplets perched on the peaks so they are shed, while the hydrophobic micropores between them stay dry and remain open to gas exchange even in the wet: the surface separates liquid transport from gas transport across the same plane (Biomimicry 3.8 and HOK 2013, pp. 41–44)	Selected. A physical answer to soiling, not a chemical one
Desert shrub <i>Encelia farinosa</i>	F2, F3	Dense leaf pubescence raises reflectance and is adjusted seasonally (Ehleringer and Mooney 1978)	Selected as the biological precedent for a seasonally switching surface
<i>De-selected, with reasons</i>			
Termite mound	F5	The classic reference, but the “chimney air-conditioning” account is contested; measurements support wind-driven transient gas exchange rather than a thermosiphon (Turner and Soar 2008)	De-selected. The evidence does not support the mechanism, and the building usually cited for it uses fans
Namib desert beetle	—	Hydrophilic–hydrophobic patterning harvests water from fog	De-selected twice over: fog in Zurich is a winter phenomenon and a heat wave brings neither rain nor fog, and the water path is outside the boundary (Table 4)
Thomson’s gazelle carotid rete	F5	Counter-current heat exchange cools the brain	De-selected as a whole-organism model — internal physiology. The counter-current <i>arrangement</i> is retained as an option for the liquid loop in Section 3.2
Tenrec aestivation	—	Dormancy through the hot season	De-selected. A behaviour; an envelope has none
Green façades and roofs	F2, F3	Shading and evapotranspiration from planting	Out of scope by the project boundary (Table 4), and in direct competition with our own water reserve: planting needs most water exactly when least is available

B.2 Decision log

Every decision below is argued at the place named in the last column. This table exists so that the reasoning can be checked without re-reading the report, and so that what was *not* chosen stays visible: a design process is only legible if the discarded alternatives are on the record.

Table B.1: Decision log. What was chosen, what it displaced, and why.

Decision	Chosen	Rejected alternative	Reason, and where it is argued
Design question	Let the absorbed heat drive its own removal, self-regulating	Three other candidates, from “prevent heat build-up in cities” to “prevent heat waves causing heat islands”	Too broad, unanchored, or factually wrong. The chosen question adds the self-regulating clause, which is the actual content of the project. Section 2.1
Reference city	Zurich	A generic European city	An adopted municipal heat strategy, published target values, open geodata and long-term stations make the targets checkable instead of invented. Section 2.2
Reference site	Modellierungsgebiet 03, closed perimeter block	A freely chosen building	The city publishes modelled baseline and climate-optimised results for exactly this area, so the concept is compared against a documented reference. Section 2.2
Design case	Five consecutive hot days	One hot afternoon	Only a multi-day case stresses the water reserve and the reset behaviour of the store; a concept that works on day one and saturates by day three has not solved the problem. Section 2.2
Evening window	18:00–24:00	20:00–24:00	A window must contain the instant at which its own KPI is evaluated, and the two hours before 20:00 are among the hottest of the day at the surface. Section 2.2
System of interest	The façade, with the roof as the release interface	The envelope as a whole, roof included	The roof receives more than twice the noon irradiance, but releases its heat upward to a sky it can see; the façade releases sideways into the canyon at head height, and Section 1 fixes that as where the effect must appear. Section 2.2
Number of systems	One	Two — “reduce heat” and “prevent heat”	The systems view showed these are two functions of one system, coupled by the self-regulation decision. Section 2.3

Decision	Chosen	Rejected alternative	Reason, and where it is argued
Scope of “use the heat”	Let it drive its own removal	Seasonal storage and electricity generation	Conversion competes with photovoltaics for the same surface and any seasonal store lies outside the envelope. Section 2.3
Selection KPI for the urban effect	PET at 2 m above ground	Urban heat island intensity	A district-scale quantity that a single retrofit cannot move measurably, and it invites the surface-versus-air confusion. Section 2.6
Cost and embodied emissions	Constraints	A seventh KPI	A KPI must follow from a function; these are conditions to satisfy, not measures of performance. The move also keeps the list inside the required four to seven. Section 2.6
Albedo target	≥ 0.80 , the city’s threshold	The technical maximum of ≈ 0.98	Life’s Principles ask for optimised rather than maximised; further reflectance costs glare, winter gains and photovoltaic yield. Section 2.6
Life’s Principles claimed	Three used as scored criteria, two as rules	“Creates opportunities for life” and “Evolve to survive” for the artefact	A damp textured surface invites the algal growth KPI 6 penalises, and a façade layer does not evolve across generations. Naming the conflict is more honest than claiming the principle. Section 2.6
Self-cleaning	Physical: multi-scale surface geometry	A biocide or film stabiliser, the obvious answer to soiling	Whatever leaves the outer finish is abraded into the street canyon the project exists to improve, so the life-friendly-chemistry principle rules it out before any concept exists. This is why the lichen entered the model table at all. Section 4.7
Search strategy	By function, through the taxonomy	By organism	Searching for organisms would not have produced F7, which is the pairing of two taxonomy branches rather than a single one. Section 3.1.1
Termite mound	De-selected	The standard biomimicry reference for passive cooling	The chimney account is contested in the literature, and the building usually cited for it uses fans. Section 3.1
Fog harvesting	De-selected	The Namib beetle, the most-cited water model	Fog in Zurich is a winter phenomenon; a heat wave brings neither rain nor fog. Section 3.1

Decision	Chosen	Rejected alternative	Reason, and where it is argued
Green façades	Out of scope	An obvious and proven measure	Already the client's HA 09 and HA 10, and in direct competition with our own water reserve, which is scarcest exactly when planting needs most. Section 3.1
Evidence standard	Primary papers where they exist, the publishing organisation otherwise	AskNature entries treated as evidence in their own right	AskNature and AI-assisted search find candidates. Ten of twenty-one model entries still rest on an AskNature page alone; this is recorded as a shortfall rather than presented as met. Section 3.1.1
Water path	Out of the boundary; the concept is dry	Capture, retention and metered evaporation of rainwater (F6–F8 and a water KPI in the earlier scoping)	Evaporation is the steepest load-following mechanism, but the reserve must be charged before a rainless heat wave, the damp layer invites the algal growth KPI 6 penalises, and it must survive twenty freeze–thaw cycles. Section 2.3; the cost of the decision is stated in Section 4
Concept pair	[TODO: decide]	[TODO: the two directions not developed]	[TODO: state the physical distinction that makes the pair meaningful]
Final concept	[TODO: A, B or a justified hybrid]	[TODO:]	[TODO: name the weakness in V1 that V2 answers]